EVIDENCIAS PARCIAL 1 - SERVICIOS TELEMATICOS

Estudiante: Juan Camilo Ballesteros Sierra
Codigo: 2230721
Profesor: Oscar Hernan Mondragon Martinez
Dominio configurado: parcial.juan-camilo.com
Fecha de generacion: 2026-03-03 19:01:40

URL publica de ngrok:

```
■URL publica ngrok (capturada):
https://rapaciously-slinkier-annabelle.ngrok-free.dev
Fecha: 2026-03-03 17:57:45
```

Punto 1 - DNS Maestro/Esclavo

```
■EVIDENCIA PUNTO 1 - DNS Maestro/Esclavo
Fecha: 2026-03-03 15:02:56

=== MASTER /etc/bind/named.conf.options ===
options {
    directory "/var/cache/bind";

    recursion no;
    allow-recursion { none; };
    allow-query { any; };
    allow-transfer { none; };

    dnssec-validation auto;
    listen-on-v6 { any; };
};

=== MASTER /etc/bind/named.conf.local ===
key "axfr-key" {
    algorithm hmac-sha256;
    secret "/A39HXgLEpVmfTRAlsn+FhjExOzpZjbj/r/eanLR9PU=";
};

zone "empresa.local" {
    type master;
    file "/etc/bind/db.empresa.local";
    notify yes;
    allow-transfer { key "axfr-key"; };
};

zone "50.168.192.in-addr.arpa" {
    type master;
    file "/etc/bind/db.192.168.50";
    notify yes;
    allow-transfer { key "axfr-key"; };
};

=== MASTER db.empresa.local ===
$TTL 86400
@   IN  SOA maestro.empresa.local. admin.empresa.local. (
        2026030302 ; Serial
        3600       ; Refresh
        1800       ; Retry
        604800     ; Expire
        86400 )    ; Minimum TTL

@                   IN  NS      maestro.empresa.local.
@                   IN  NS      esclavo.empresa.local.

maestro             IN  A       192.168.50.3
maestro             IN  AAAA    fd00:50::3
esclavo             IN  A       192.168.50.2
esclavo             IN  AAAA    fd00:50::2
www                 IN  CNAME   maestro
app                 IN  CNAME   esclavo
api2                IN  A       192.168.50.10
=== MASTER db.192.168.50 ===
```

```
$TTL 86400
@   IN  SOA maestro.empresa.local. admin.empresa.local. (
        2026030301 ; Serial
        3600
        1800
        604800
        86400 )

@       IN  NS      maestro.empresa.local.
@       IN  NS      esclavo.empresa.local.

3       IN  PTR     maestro.empresa.local.
2       IN  PTR     esclavo.empresa.local.

=== SLAVE /etc/bind/named.conf.options ===
options {
    directory "/var/cache/bind";

    recursion no;
    allow-recursion { none; };
    allow-query { any; };
    allow-transfer { none; };

    dnssec-validation auto;
    listen-on-v6 { any; };
};

=== SLAVE /etc/bind/named.conf.local ===
key "axfr-key" {
    algorithm hmac-sha256;
    secret "/A39HXgLEpVmfTRAlsn+FhjExOzpZjbj/r/eanLR9PU=";
};

zone "empresa.local" {
    type slave;
    file "/var/cache/bind/db.empresa.local";
    masters { 192.168.50.3 key "axfr-key"; };
};

zone "50.168.192.in-addr.arpa" {
    type slave;
    file "/var/cache/bind/db.192.168.50";
    masters { 192.168.50.3 key "axfr-key"; };
};

=== PRUEBAS DIG DESDE ESCLAVO ===
192.168.50.3
192.168.50.2
maestro.empresa.local.
192.168.50.3
fd00:50::3
maestro.empresa.local.
esclavo.empresa.local.
maestro.empresa.local. admin.empresa.local. 2026030302 3600 1800 604800 86400
192.168.50.10

=== AXFR NO AUTORIZADO (debe fallar) ===

; <<>> DiG 9.18.39-0ubuntu0.22.04.2-Ubuntu <<>> @192.168.50.3 empresa.local AXFR +time=2 +tries=1
; (1 server found)
;; global options: +cmd
; Transfer failed.

=== RECURSION DESHABILITADA ===

=== LOGS DE TRANSFERENCIA/NOTIFY EN ESCLAVO ===
Mar 03 19:57:09 cliente named[2599]: zone empresa.local/IN: Transfer started.
Mar 03 19:57:09 cliente named[2599]: transfer of 'empresa.local/IN' from 192.168.50.3#53: connected using
192.168.50.3#53 TSIG axfr-key
Mar 03 19:57:09 cliente named[2599]: zone empresa.local/IN: transferred serial 2026030301: TSIG 'axfr-key'
Mar 03 19:57:09 cliente named[2599]: transfer of 'empresa.local/IN' from 192.168.50.3#53: Transfer status: succes
Mar 03 19:57:09 cliente named[2599]: transfer of 'empresa.local/IN' from 192.168.50.3#53: Transfer completed: 1
messages, 10 records, 358 bytes, 0.001 secs (358000 bytes/sec) (serial 2026030301)
Mar 03 19:57:09 cliente named[2599]: zone empresa.local/IN: sending notifies (serial 2026030301)
Mar 03 19:57:10 cliente named[2599]: zone 50.168.192.in-addr.arpa/IN: Transfer started.
Mar 03 19:57:10 cliente named[2599]: transfer of '50.168.192.in-addr.arpa/IN' from 192.168.50.3#53: connected usi
192.168.50.3#53 TSIG axfr-key
Mar 03 19:57:10 cliente named[2599]: zone 50.168.192.in-addr.arpa/IN: transferred serial 2026030301: TSIG 'axfr-k
Mar 03 19:57:10 cliente named[2599]: transfer of '50.168.192.in-addr.arpa/IN' from 192.168.50.3#53: Transfer stat
success
```

```
Mar 03 19:57:10 cliente named[2599]: transfer of '50.168.192.in-addr.arpa/IN' from 192.168.50.3#53: Transfer comp
1 messages, 6 records, 289 bytes, 0.001 secs (289000 bytes/sec) (serial 2026030301)
Mar 03 19:57:10 cliente named[2599]: zone 50.168.192.in-addr.arpa/IN: sending notifies (serial 2026030301)
Mar 03 19:58:23 cliente named[2599]: client @0x7f2b00010488 192.168.50.3#43882: received notify for zone 'empresa
Mar 03 19:58:23 cliente named[2599]: zone empresa.local/IN: notify from 192.168.50.3#43882: serial 2026030302
Mar 03 19:58:23 cliente named[2599]: zone empresa.local/IN: Transfer started.
Mar 03 19:58:23 cliente named[2599]: transfer of 'empresa.local/IN' from 192.168.50.3#53: connected using
192.168.50.3#53 TSIG axfr-key
Mar 03 19:58:23 cliente named[2599]: zone empresa.local/IN: transferred serial 2026030302: TSIG 'axfr-key'
Mar 03 19:58:23 cliente named[2599]: transfer of 'empresa.local/IN' from 192.168.50.3#53: Transfer status: succes
Mar 03 19:58:23 cliente named[2599]: transfer of 'empresa.local/IN' from 192.168.50.3#53: Transfer completed: 1
messages, 11 records, 379 bytes, 0.001 secs (379000 bytes/sec) (serial 2026030302)
Mar 03 19:58:23 cliente named[2599]: zone empresa.local/IN: sending notifies (serial 2026030302)
Mar 03 19:58:23 cliente named[2599]: client @0x7f2b00010488 192.168.50.3#39920: received notify for zone 'empresa
Mar 03 19:58:23 cliente named[2599]: zone empresa.local/IN: notify from 192.168.50.3#39920: zone is up to date
Mar 03 20:00:46 cliente named[2599]: client @0x7f2b00010488 192.168.50.3#45626: received notify for zone '50.168.
addr.arpa'
Mar 03 20:00:46 cliente named[2599]: zone 50.168.192.in-addr.arpa/IN: notify from 192.168.50.3#45626: zone is up
Mar 03 20:00:47 cliente named[2599]: client @0x7f2afd2bba08 192.168.50.3#59875: received notify for zone 'empresa
Mar 03 20:00:47 cliente named[2599]: zone empresa.local/IN: notify from 192.168.50.3#59875: zone is up to date
Mar 03 20:01:58 cliente named[2599]: client @0x7f2b00010488 192.168.50.3#57819: received notify for zone 'empresa
Mar 03 20:01:58 cliente named[2599]: zone empresa.local/IN: notify from 192.168.50.3#57819: zone is up to date
Mar 03 20:01:59 cliente named[2599]: client @0x7f2afd2bba08 192.168.50.3#45460: received notify for zone '50.168.
addr.arpa'
Mar 03 20:01:59 cliente named[2599]: zone 50.168.192.in-addr.arpa/IN: notify from 192.168.50.3#45460: zone is up
Mar 03 20:02:15 cliente named[3384]: zone 50.168.192.in-addr.arpa/IN: loaded serial 2026030301
Mar 03 20:02:15 cliente named[3384]: zone empresa.local/IN: loaded serial 2026030302
Mar 03 20:02:15 cliente named[3384]: zone empresa.local/IN: sending notifies (serial 2026030302)
Mar 03 20:02:15 cliente named[3384]: zone 50.168.192.in-addr.arpa/IN: sending notifies (serial 2026030301)
Mar 03 20:02:25 cliente named[3384]: client @0x7f53452b6f18 192.168.50.3#38662: received notify for zone 'empresa
Mar 03 20:02:25 cliente named[3384]: zone empresa.local/IN: notify from 192.168.50.3#38662: zone is up to date
Mar 03 20:02:25 cliente named[3384]: client @0x7f534c009928 192.168.50.3#36178: received notify for zone '50.168.
addr.arpa'
Mar 03 20:02:25 cliente named[3384]: zone 50.168.192.in-addr.arpa/IN: notify from 192.168.50.3#36178: zone is up
Mar 03 20:02:26 cliente named[3384]: client @0x7f53452b6f18 192.168.50.3#36523: received notify for zone 'empresa
Mar 03 20:02:26 cliente named[3384]: zone empresa.local/IN: notify from 192.168.50.3#36523: zone is up to date


=== PRUEBA DE FALLA DEL MAESTRO ===
Apagando maestro...
==> servidor: Forcing shutdown of VM...
Consultas al esclavo con maestro apagado:
192.168.50.3
192.168.50.10
maestro.empresa.local.
Encendiendo maestro nuevamente...
Bringing machine 'servidor' up with 'virtualbox' provider...
==> servidor: Checking if box 'bento/ubuntu-22.04' version '202510.26.0' is up to date...
==> servidor: Clearing any previously set forwarded ports...
==> servidor: Clearing any previously set network interfaces...
==> servidor: Preparing network interfaces based on configuration...
    servidor: Adapter 1: nat
    servidor: Adapter 2: hostonly
==> servidor: Forwarding ports...
    servidor: 22 (guest) => 2222 (host) (adapter 1)
==> servidor: Booting VM...
==> servidor: Waiting for machine to boot. This may take a few minutes...
    servidor: SSH address: 127.0.0.1:2222
    servidor: SSH username: vagrant
    servidor: SSH auth method: private key
    servidor: Warning: Connection reset. Retrying...
    servidor: Warning: Connection aborted. Retrying...
==> servidor: Machine booted and ready!
==> servidor: Checking for guest additions in VM...
==> servidor: Setting hostname...
==> servidor: Configuring and enabling network interfaces...
==> servidor: Mounting shared folders...
    servidor: C:/Users/USUARIO/Desktop/UAO 2026 SEMESTRE 1/Servicios telematicos/prueba => /vagrant
==> servidor: Machine not provisioned because `--no-provision` is specified.
Estado final de VMs:
Current machine states:

servidor                  running (virtualbox)
cliente                   running (virtualbox)

This environment represents multiple VMs. The VMs are all listed
above with their current state. For more information about a specific
VM, run `vagrant status NAME`.

=== AXFR NO AUTORIZADO CONTRA ESCLAVO (debe fallar) ===
```

```
; <<>> DiG 9.18.39-0ubuntu0.22.04.2-Ubuntu <<>> @192.168.50.2 empresa.local AXFR +time=2 +tries=1
; (1 server found)
;; global options: +cmd
; Transfer failed.
```

Punto 2 - Apache + Compresion

```
■EVIDENCIA FINAL PUNTO 2 - APACHE + MOD_DEFLATE + DNS
Dominio usado: parcial.juan-camilo.com
Fecha: 2026-03-03 17:55:11

=== 1) DNS LOCAL DEL DOMINIO ===
192.168.50.3

=== 2) MODULOS APACHE ACTIVOS ===
 deflate_module (shared)
 headers_module (shared)

=== 3) VHOST PARCIAL ===
<VirtualHost *:80>
    ServerName parcial.juan-camilo.com
    ServerAdmin webmaster@localhost
    DocumentRoot /var/www/parcial

    <Directory /var/www/parcial>
        AllowOverride None
        Require all granted
    </Directory>

    ErrorLog ${APACHE_LOG_DIR}/parcial_error.log
    CustomLog ${APACHE_LOG_DIR}/parcial_access.log combined
</VirtualHost>

=== 4) CONFIGURACION DE COMPRESION ===
<IfModule mod_deflate.c>
    AddOutputFilterByType DEFLATE text/html text/plain text/css text/javascript application/javascript applicatio
application/xml text/xml image/svg+xml
</IfModule>

<IfModule mod_headers.c>
    Header append Vary Accept-Encoding
</IfModule>

# Exclude already compressed or binary-heavy formats.
SetEnvIfNoCase Request_URI "\.(?:gif|jpe?g|png|webp|avif|mp4|avi|mkv|mov|zip|gz|bz2|7z|rar|pdf)$" no-gzip dont-va

=== 5) CABECERAS CON GZIP (texto) ===
HTTP/1.1 200 OK

Date: Tue, 03 Mar 2026 22:56:13 GMT

Server: Apache/2.4.52 (Ubuntu)

Last-Modified: Tue, 03 Mar 2026 22:53:24 GMT

ETag: "b5a40-64c269224a6cd-gzip"

Accept-Ranges: bytes

Vary: Accept-Encoding

Content-Encoding: gzip

Content-Length: 2258

Content-Type: text/plain



=== 6) CABECERAS SIN GZIP (imagen excluida) ===
HTTP/1.1 200 OK

Date: Tue, 03 Mar 2026 22:56:17 GMT

Server: Apache/2.4.52 (Ubuntu)

Last-Modified: Tue, 03 Mar 2026 22:53:24 GMT

ETag: "a000-64c2692246855"

Accept-Ranges: bytes
```

```
Content-Length: 40960

Vary: Accept-Encoding

Content-Type: image/png



=== 7) COMPARACION DE TAMANO ===
plain_bytes=744000
gzip_bytes=2258

=== 8) SNIFFER (TSHARK) ===
Running as user "root" and group "root". This could be dangerous.

200             29012
200     gzip    2609

=== 9) ARCHIVOS DEL SITIO ===
total 784K
-rw-r--r-- 1 root root  116 Mar  3 22:53 app.js
-rw-r--r-- 1 root root  394 Mar  3 22:53 index.html
-rw-r--r-- 1 root root  40K Mar  3 22:53 logo.png
-rw-r--r-- 1 root root 727K Mar  3 22:53 lorem.txt
-rw-r--r-- 1 root root  332 Mar  3 22:53 pagina_personalizada.html
-rw-r--r-- 1 root root  228 Mar  3 22:53 styles.css
```

Evidencia Visual Punto 2

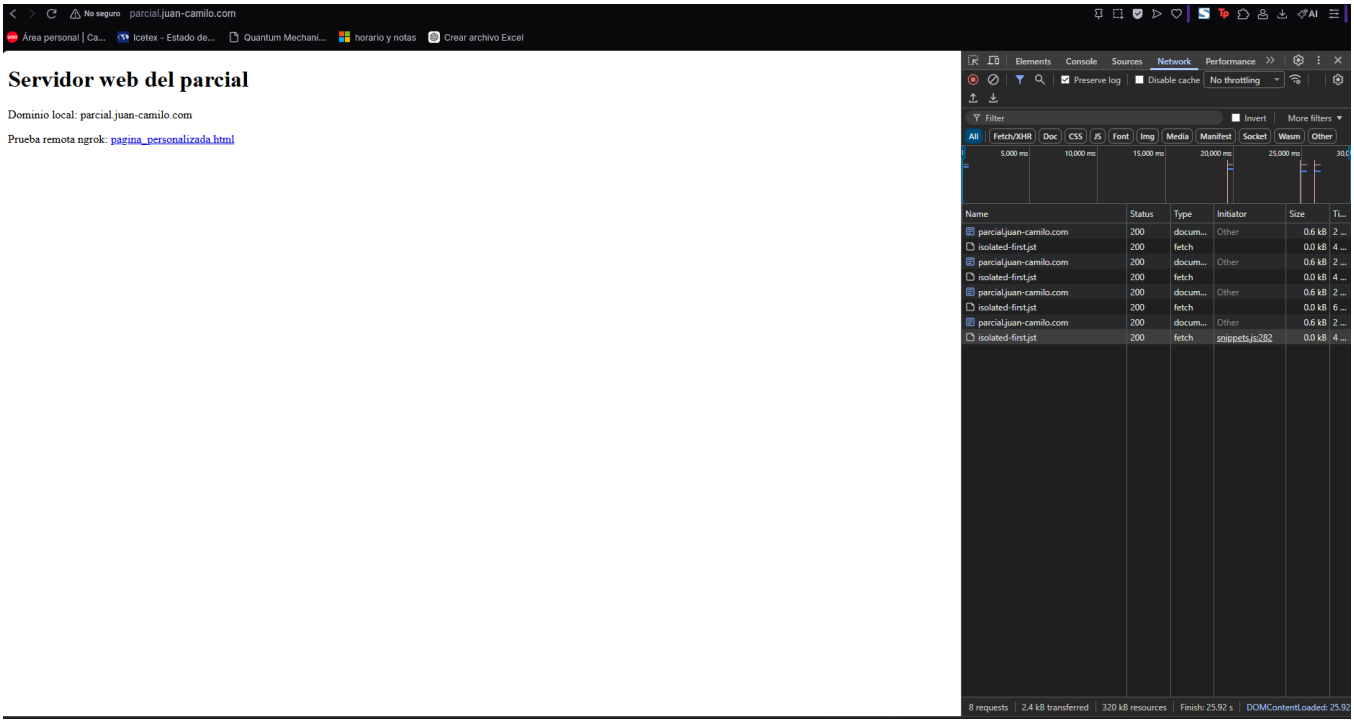

Figura 1. Vista de sitio y panel Network.

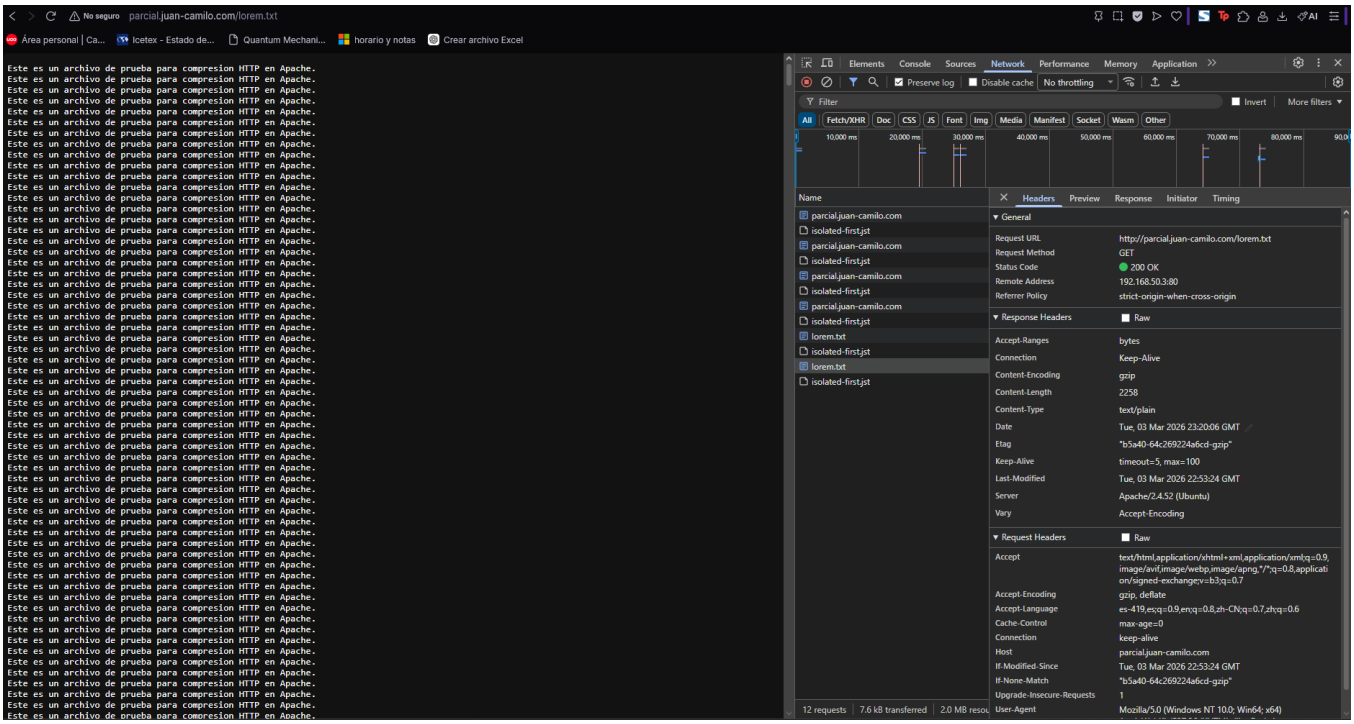

Figura 2. lorem.txt con Content-Encoding: gzip.

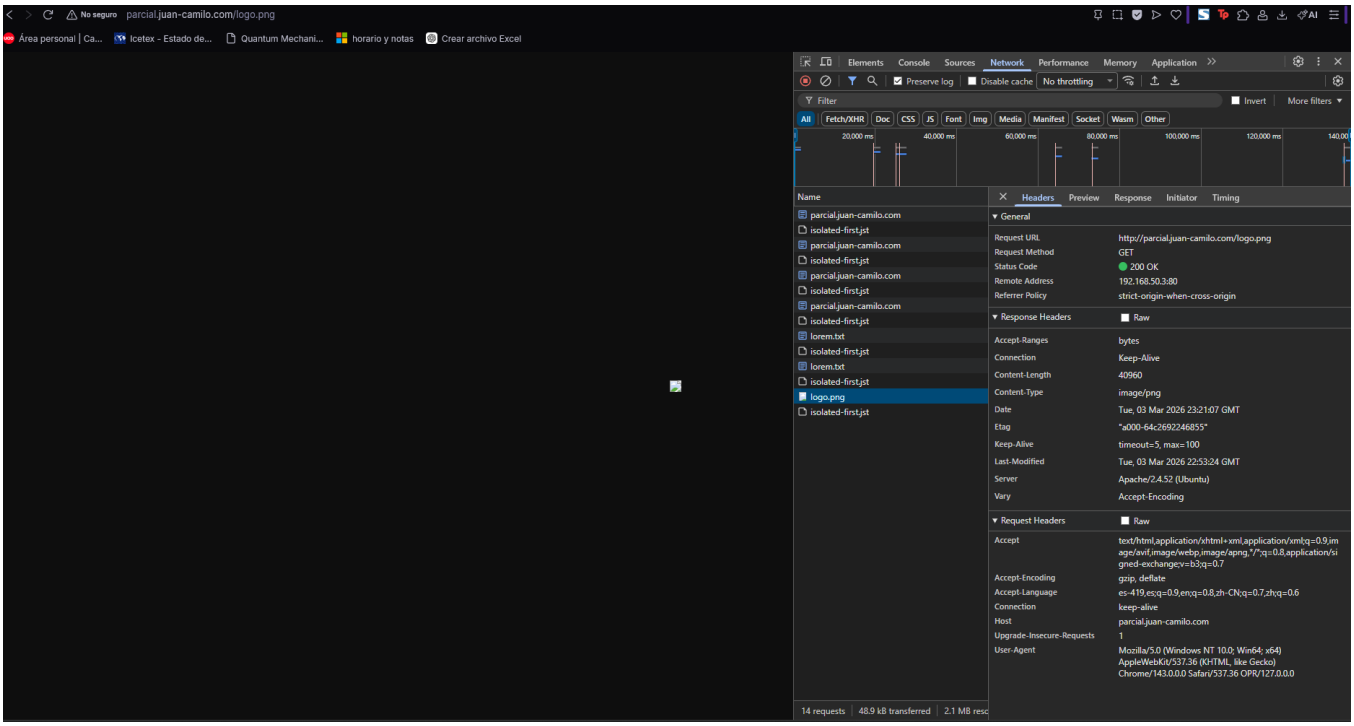

Figura 3. logo.png sin Content-Encoding: gzip.

Evidencia Wireshark Punto 2

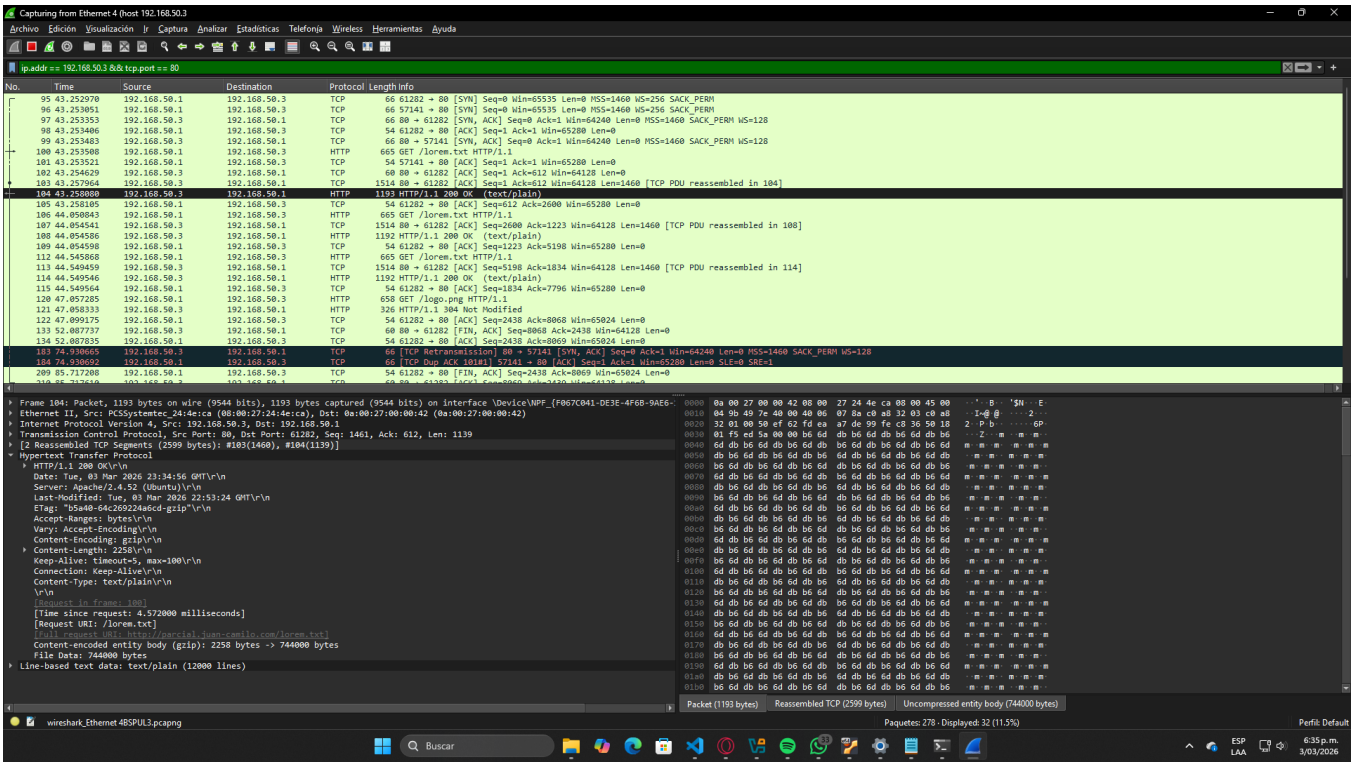

Figura 4. Wireshark: respuesta HTTP 200 de lorem.txt con Content-Encoding: gzip.

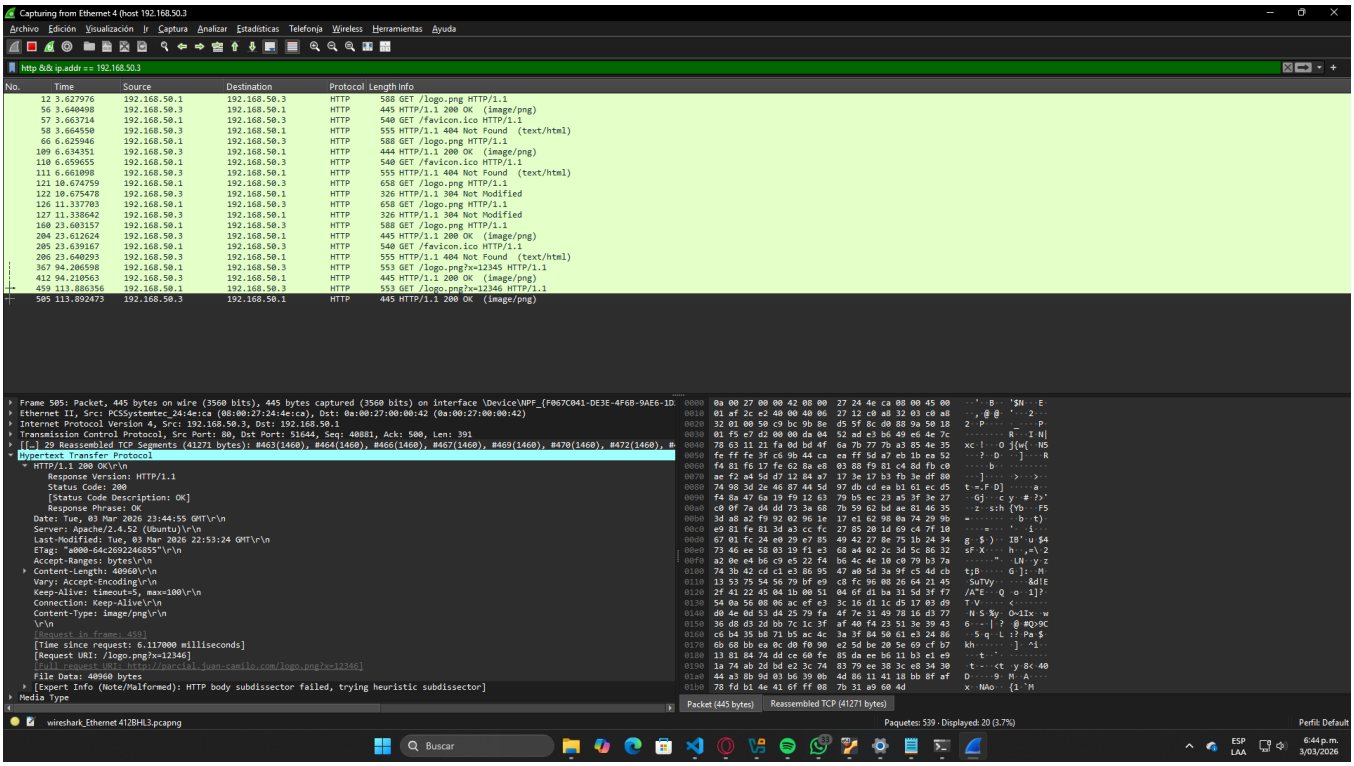

Figura 5. Wireshark: respuesta HTTP 200 de logo.png sin Content-Encoding.

Punto 3 - Ngrok

```
■EVIDENCIA FINAL PUNTO 3 - NGROK + PAGINA PERSONALIZADA
Dominio local del sitio: parcial.juan-camilo.com
Fecha: 2026-03-03 17:55:57

=== SITIOS HABILITADOS ===
total 0
lrwxrwxrwx 1 root root 31 Mar  3 20:14 parcial.conf -> ../sites-available/parcial.conf

=== VERSION NGROK ===
ngrok version 3.36.1

=== URL PUBLICA ===
https://rapaciously-slinkier-annabelle.ngrok-free.dev

=== VERIFICACION REMOTA ===
Probando URL publica: https://rapaciously-slinkier-annabelle.ngrok-free.dev
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed

  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
  0   332    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
HTTP/2 200

accept-ranges: bytes

content-type: text/html

date: Tue, 03 Mar 2026 22:56:56 GMT

etag: "14c-64c2692a119dd"

last-modified: Tue, 03 Mar 2026 22:53:32 GMT

ngrok-agent-ips: 191.111.253.149

server: Apache/2.4.52 (Ubuntu)

vary: Accept-Encoding

content-length: 332



<!doctype html>
<html lang="es">
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Pagina Personalizada Parcial</title>
</head>
<body>
  <h1>Parcial Servicios Telematicos - Punto 3</h1>
  <p>Esta es la pagina personalizada publicada por tunel ngrok.</p>
```